

4.2–4.3 Projections and Least Squares

Concise exam revision notes for projections, projection matrices, and least squares.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. When $Ax = b$ has no solution, replace b by the closest vector p in the column space of A . The error $e = b - p$ must be perpendicular to that column space.

The picture is always

$$b = p + e, \quad p \in C(A), \quad e \in N(A^T).$$

Projection turns an unsolvable equation into a solvable normal equation.

2 Projection Onto a Line

For a nonzero vector a , the projection of b onto the line through a has the form

$$p = \hat{x}a.$$

The error is perpendicular to a :

$$a^T(b - \hat{x}a) = 0.$$

Solving gives

$$\hat{x} = \frac{a^T b}{a^T a}, \quad p = a \frac{a^T b}{a^T a}.$$

So the one-dimensional projection matrix is

$$P = \frac{aa^T}{a^T a}.$$

Meaning. $\hat{x}$ is the coordinate along the line; p is the closest point on the line.

3 Projection Onto a Subspace

Let the columns of A span the subspace. We look for

$$p = A\hat{x}.$$

The error $e = b - A\hat{x}$ must be perpendicular to every column of A :

$$A^T e = 0.$$

Therefore

$$A^T(b - A\hat{x}) = 0, \quad A^T A\hat{x} = A^T b.$$

These are the normal equations.

If the columns of A are independent, then $A^T A$ is invertible and

$$\hat{x} = (A^T A)^{-1} A^T b.$$

The projection is

$$p = A\hat{x} = A(A^T A)^{-1} A^T b.$$

So the projection matrix is

$$P = A(A^T A)^{-1} A^T.$$

4 Projection Matrix Facts

Fact	Formula	Meaning
Projection lands in $C(A)$	$p = Pb$	p is the closest vector in the column space.
Error is perpendicular	$A^T(b - p) = 0$	$e = b - p$ lies in $N(A^T)$.
Projecting twice changes nothing	$P^2 = P$	Once on the subspace, it stays there.
Orthogonal projection is symmetric	$P^T = P$	True for projection onto a column space.

Exam trigger. If the problem asks for closest vector, best fit, or least squares, expect $A^T A\hat{x} = A^T b$.

5 Orthogonal Columns Shortcut

If the columns $q_1, \dots, q_n$ are orthonormal, put them in Q . Then

$$Q^T Q = I.$$

Projection becomes much simpler:

$$p = Q Q^T b.$$

Equivalently,

$$p = (q_1^T b)q_1 + \dots + (q_n^T b)q_n.$$

Columns of A	Projection formula
Independent	$P = A(A^T A)^{-1} A^T$.
Orthogonal but not unit	Add projections along each column separately.
Orthonormal	$P = Q Q^T$.

Common mistake. Do not use $Q Q^T$ unless the columns are orthonormal. Orthogonal columns still need length factors.

6 Least Squares

Least squares solves the closest version of an inconsistent system:

$$Ax \approx b.$$

The best coefficient vector is $\hat{x}$, found from

$$A^T A \hat{x} = A^T b.$$

Then

$$p = A \hat{x}, \quad e = b - p.$$

For a best-fit line

$$y = C + Dt,$$

the matrix has rows

$$\begin{bmatrix} 1 & t_i \end{bmatrix}.$$

The unknown vector is

$$\hat{x} = \begin{bmatrix} C \\ D \end{bmatrix}.$$

The same normal equation gives the least-squares line.

7 P and $I - P$

The projection splits b into two perpendicular parts:

$$b = Pb + (I - P)b.$$

Here

$$Pb \in C(A), \quad (I - P)b \in N(A^T).$$

If P projects onto $C(A)$, then $I - P$ projects onto the left nullspace.

Exam memory. Projection onto a subspace plus projection onto its orthogonal complement gives the original vector.

8 Common Mistakes

Mistake	Correct exam habit
Solving $Ax = b$ when it is inconsistent.	Solve $A^T A \hat{x} = A^T b$.
Forgetting the error condition.	Check $A^T e = 0$.
Using $P = A^{-1}$ language.	Projection is not inversion. Use $P = A(A^T A)^{-1} A^T$.
Using $Q Q^T$ for non-orthonormal columns.	First check $Q^T Q = I$.
Assuming $P_1 + P_2 = I$ for any two projections.	This works for orthogonal complementary subspaces.

9 Final Exam Checklist

Checklist.

1. Identify the subspace: usually $C(A)$ or a line through a .
2. Write $p = A\hat{x}$ or $p = \hat{x}a$.
3. Write the error $e = b - p$.
4. Impose perpendicularity: $A^T e = 0$ or $a^T e = 0$.
5. Solve $A^T A \hat{x} = A^T b$.
6. Compute $p = A\hat{x}$.
7. Compute the error $e = b - p$.
8. Use $P = A(A^T A)^{-1} A^T$ only when the columns of A are independent.
9. Use $P = Q Q^T$ only when the columns are orthonormal.
10. State where p and e live: $p \in C(A)$, $e \in N(A^T)$.